TMUA PRACTICE TEST SERIES

TMUA Topic Test 3

Calculus

- 20 multiple-choice questions
- Recommended time: 75 minutes
- Non-calculator
- Paper 1 styled topic revision set

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

TMUA Topic Test 3 — Calculus

Complete, independently checked worked solutions for the 20-question TMUA Paper 1 styled calculus topic test.

TMUA Topic Test 3 answer key

Q1 **A**	Q2 **C**	Q3 **C**	Q4 **D**	Q5 **A**	Q6 **B**	Q7 **F**	Q8 **C**	Q9 **A**	Q10 **B**
Q11 **D**	Q12 **A**	Q13 **D**	Q14 **B**	Q15 **D**	Q16 **F**	Q17 **A**	Q18 **D**	Q19 **D**	Q20 **E**

WORKED SOLUTIONS

Integration, differentiation and optimisation

Questions 1–6 · substitution, structural differentiation, areas, increasing functions, optimisation and basic differential equations.

Q1 Integration & Substitution **Answer A**

Question 1

Compute the integral

$$\int_0^1 \frac{x-4}{\sqrt{x}+2}\,dx.$$

- **A** $-\frac{4}{3}$
- **B** -3
- **C** -1
- **D** $\frac{2}{3}$
- **E** 1

WORKED SOLUTION

Let $t=\sqrt{x}$. Then $x=t^2$, $dx=2t\,dt$, and the limits remain 0 to 1.

$$
\begin{aligned}
I &= \int_0^1 \frac{t^2-4}{t+2}\,2t\,dt \\
&= \int_0^1 2t(t-2)\,dt \\
&= \int_0^1 (2t^2-4t)\,dt \\
&= \left[\frac{2}{3}t^3 - 2t^2\right]_0^1 = -\frac{4}{3}.
\end{aligned}
$$

Therefore the answer is $\boxed{\text{A}}$.

Q2 Differentiation & Structure **Answer C**

Question 2

Given

$$f(x) = \left(9x^2 + 12 + \frac{4}{x^2}\right)^{1/2}$$

and

$$\frac{d^n f}{dx^n}(2) = -\frac{3}{4},$$

find n.

A 1 B 2 C 3 D 4 E 5

WORKED SOLUTION

The expression inside the square root is a perfect square:

$$9x^2 + 12 + \frac{4}{x^2} = \left(3x + \frac{2}{x}\right)^2.$$

Near $x = 2$, the quantity $3x + 2/x$ is positive, so

$$f(x) = 3x + \frac{2}{x}.$$

Differentiate:

$$f'(x) = 3 - \frac{2}{x^2}, \qquad f''(x) = \frac{4}{x^3}, \qquad f^{(3)}(x) = -\frac{12}{x^4}.$$

Thus

$$f^{(3)}(2) = -\frac{12}{16} = -\frac{3}{4}.$$

Hence $n = 3$, so the answer is $\boxed{\text{C}}$.

Q3 Area Between Curves **Answer C**

Question 3

For $p > 0$, find the area enclosed by the curves

$$y = px^2 \qquad \text{and} \qquad x = py^2.$$

A $\frac{1}{3p}$

B $3p$

C $\frac{1}{3p^2}$

D $\frac{1}{2}p^2$

E $\frac{1}{6p^2}$

WORKED SOLUTION

From $x = py^2$, the upper branch in the first quadrant is

$$y = \sqrt{\frac{x}{p}}.$$

At an intersection with $y = px^2$,

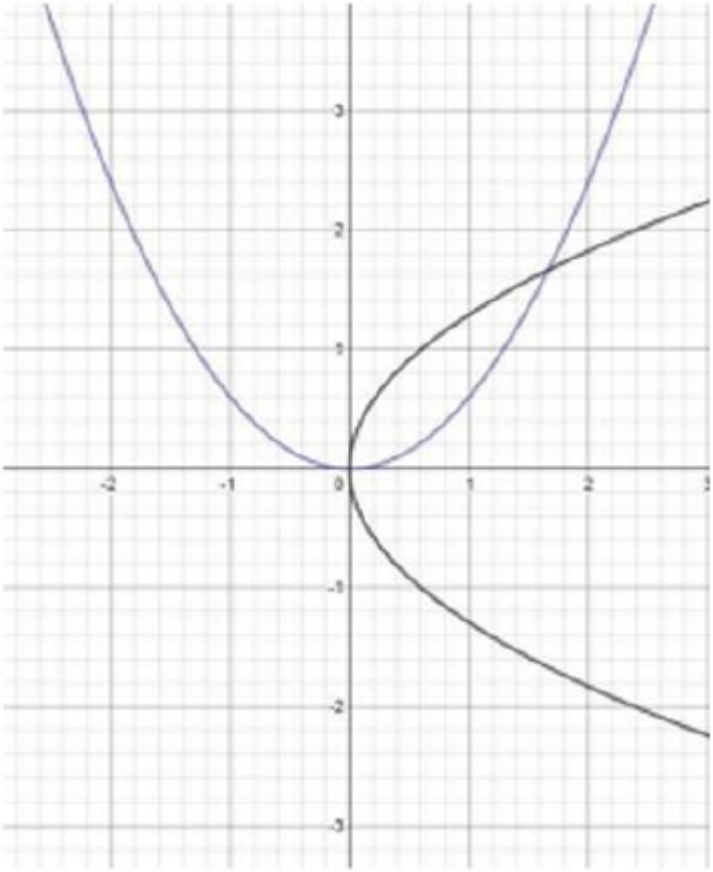
$$x = p(px^2)^2 = p^3x^4,$$

so $x = 0$ or $x = 1/p$. The corresponding non-zero point is $(1/p, 1/p)$.

Therefore the enclosed area is

$$\begin{aligned} A &= \int_0^{1/p} \left(\sqrt{\frac{x}{p}} - px^2 \right) dx \\ &= \left[\frac{2x^{3/2}}{3\sqrt{p}} - \frac{px^3}{3} \right]_0^{1/p} \\ &= \frac{2}{3p^2} - \frac{1}{3p^2} = \frac{1}{3p^2}. \end{aligned}$$

The answer is $\boxed{\text{C}}$.



Original supporting graph URL retained from the supplied solution PDF, with an embedded fallback for offline use.

Q4

Increasing Functions

Answer D

Question 4

What is the complete set of values for which

$$\frac{x^2+2x}{\sqrt{x^3}}$$

is increasing?

A $x > 3$

B $x < 2$

C $2 < x < 4$

D $x > 2$

E $0 < x < 2$

WORKED SOLUTION

The denominator requires $x > 0$. Rewrite the function as

$$x^{1/2} + 2x^{-1/2}.$$

Its derivative is

$$\frac{1}{2\sqrt{x}} - \frac{1}{x^{3/2}} = \frac{x-2}{2x^{3/2}}.$$

For $x > 0$, the denominator is positive, so the derivative is positive precisely when $x > 2$.

The answer is $\boxed{\text{D}}$.

Q5 Optimisation & Distance **Answer A**

Question 5

A point P lies on the part of the parabola $y = x^2$ in the first quadrant. What is the shortest possible distance from P to the point $(0, 3)$?

A $\frac{\sqrt{11}}{2}$

B $\frac{3}{2}$

C $\frac{\sqrt{5}}{2}$

D $\frac{11}{4}$

E $\sqrt{3}$

WORKED SOLUTION

Write $P = (x, x^2)$, where $x \geq 0$. The square of the distance to $(0, 3)$ is

$$D^2 = x^2 + (x^2 - 3)^2 = x^4 - 5x^2 + 9.$$

Minimising D is equivalent to minimising D^2. Differentiate:

$$\frac{d}{dx}(D^2) = 4x^3 - 10x = 2x(2x^2 - 5).$$

The non-zero stationary point in the first quadrant is $x^2 = 5/2$. There,

$$D^2 = \frac{5}{2} + \left(\frac{5}{2} - 3\right)^2 = \frac{5}{2} + \frac{1}{4} = \frac{11}{4}.$$

Hence

$$D_{\min} = \frac{\sqrt{11}}{2}.$$

The answer is $\boxed{\text{A}}$.

Q6

Differential Equations

Answer B

Question 6

Given that

$$\frac{dV}{dt} = (1+t)^4$$

and $V(1) = 5$, what is $V(2)$?

A $\frac{174}{5}$

B $\frac{236}{5}$

C $\frac{112}{3}$

D $\frac{89}{4}$

E $\frac{17}{32}$

WORKED SOLUTION

Use the definite change in V:

$$V(2) - V(1) = \int_1^2 (1+t)^4 \, dt = \left[\frac{(1+t)^5}{5}\right]_1^2 = \frac{3^5 - 2^5}{5} = \frac{211}{5}.$$

Therefore

$$V(2) = 5 + \frac{211}{5} = \frac{236}{5}.$$

The answer is $\boxed{\text{B}}$.

WORKED SOLUTIONS

Stationary points, symmetry and numerical methods

Questions 7–13 · parameterised stationary points, definite-integral comparisons, symmetry, turning points, the trapezium rule and cubic root conditions.

Q7 Stationary Points & Parameters **Answer F**

Question 7

Let

$$y = \left(\frac{1}{3}k^3 + 1\right)x^3 - (2k^2 + 3)x^2 + (31 - 25k)x + 43.$$

The curve has a turning point that is a maximum at $x = 1$, precisely for:

A $-4 < k < 7$	**B** $k = 1$ or 7
C $k > 7$	**D** $k = -4$ or 7
E $k < 1$	**F** $k = -4$ or 1
G $k < -4$	**H** $k = -4, 1$ or 7

WORKED SOLUTION

For a turning point at $x = 1$, require $y'(1) = 0$.

$$y'(1) = k^3 - 4k^2 - 25k + 28 = (k - 1)(k - 7)(k + 4).$$

Thus the possible values are

$$k = -4,\ 1,\ 7.$$

Now

$$y''(1) = 2k^3 - 4k^2 = 2k^2(k - 2).$$

A maximum requires $y''(1) < 0$, so among the three candidates only $k = -4$ and $k = 1$ work.

The answer is $\boxed{\text{F}}$.

Q8

Comparing Definite Integrals

Answer C

Question 8

Which of the following integrals has the largest value?

A $\displaystyle\int_0^{\pi/3} \cos\left(\frac{\pi}{2} - x\right) dx$

B $\displaystyle\int_{\pi/3}^{2\pi/3} \cos^2 x \, dx$

C $\displaystyle\int_{2\pi/3}^{\pi} -\sin\left(\frac{\pi}{2} - x\right) dx$

D $\displaystyle\int_0^{\pi} \sin\left(\frac{\pi}{2} - x\right) dx$

WORKED SOLUTION

Simplify each integrand:

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x, \qquad -\sin\left(\frac{\pi}{2} - x\right) = -\cos x.$$

Therefore

$$A = \int_0^{\pi/3} \sin x \, dx = \frac{1}{2},$$

$$B = \int_{\pi/3}^{2\pi/3} \cos^2 x \, dx = \frac{\pi}{6} - \frac{\sqrt{3}}{4},$$

$$C = \int_{2\pi/3}^{\pi} -\cos x \, dx = \frac{\sqrt{3}}{2}, \qquad D = \int_0^{\pi} \cos x \, dx = 0.$$

Since $\sqrt{3}/2 > 1/2$, and B is much smaller than $1/2$, the largest value is C.

The answer is $\boxed{\text{C}}$.

Q9

Symmetry of Integrals

Answer A

Question 9

Find the value of

$$\int_{-1}^{1} \frac{x(x^2-2)^2}{x^2+1} \, dx.$$

A 0

B 1

C 2

D 3

E $\frac{3}{7}$

F 12

WORKED SOLUTION

Let

$$f(x) = \frac{x(x^2-2)^2}{x^2+1}.$$

The numerator contains one factor of x multiplied by a function of x^2, while the denominator is also a function of x^2. Hence

$$f(-x) = -f(x).$$

So f is odd and continuous on $[-1,1]$. Therefore

$$\int_{-1}^{1} f(x)\,dx = 0.$$

The answer is $\boxed{\text{A}}$.

Q10

Turning Points & Distance

Answer B

Question 10

A curve has equation

$$y = (2a - x)(2a + x),$$

and a second curve has equation

$$y = (2a - x)(3a + x),$$

where $a \neq 0$. The distance between the maximum points of the two curves is $\sqrt{10}/3$. Find the possible values of a.

A 0

B $\pm\frac{2}{3}$

C $\pm\frac{1}{2}$

D $\pm\frac{\sqrt{10}}{3}$

E ± 1

F $\pm\frac{2\sqrt{5}}{5}$

WORKED SOLUTION

The first curve is

$$y = 4a^2 - x^2,$$

so its maximum point is

$$P = (0, 4a^2).$$

The second curve is

$$y = -x^2 - ax + 6a^2,$$

whose maximum occurs at $x = -a/2$. Substitution gives

$$Q = \left(-\frac{a}{2}, \frac{25a^2}{4}\right).$$

Hence

$$PQ^2 = \left(\frac{a}{2}\right)^2 + \left(\frac{9a^2}{4}\right)^2 = \frac{a^2}{4} + \frac{81a^4}{16}.$$

Since $PQ = \sqrt{10}/3$,

$$\frac{a^2}{4} + \frac{81a^4}{16} = \frac{10}{9}.$$

Let $u = a^2$. Then

$$729u^2 + 36u - 160 = 0 = (9u - 4)(81u + 40).$$

As $u \geq 0$, $u = 4/9$. Therefore

$$a = \pm\frac{2}{3}.$$

The answer is $\boxed{\text{B}}$.

Q11

Properties of Definite Integrals

Answer D

Question 11

A function f satisfies

$$\int_0^4 f(x)\,dx = 10, \qquad \int_{-1}^4 f(x)\,dx = 13, \qquad \int_{-4}^3 f(-x)\,dx = 2.$$

Find

$$\int_{-3}^0 f(x)\,dx.$$

A 2

B 3

C 5

D -8

E 11

F 12

WORKED SOLUTION

First,

$$\int_{-1}^{0} f(x)\,dx = \int_{-1}^{4} f(x)\,dx - \int_{0}^{4} f(x)\,dx = 13 - 10 = 3.$$

For the third integral, use $u = -x$. As x runs from -4 to 3, u runs from 4 to -3, so

$$\int_{-4}^{3} f(-x)\,dx = \int_{-3}^{4} f(u)\,du = 2.$$

Therefore

$$\int_{-3}^{0} f(x)\,dx = \int_{-3}^{4} f(x)\,dx - \int_{0}^{4} f(x)\,dx = 2 - 10 = -8.$$

The answer is $\boxed{\text{D}}$.

Topic Test 3 · Paper 1

Question 12

A pump's discharge rate, in litres per minute, is measured at times $t = 0, 2, 4$ minutes. The readings are $3, 9, 5$, respectively. A fourth reading is found at $t = 3$. Using all four readings, the trapezium estimate of the total discharge is 28.5 litres. What rate was measured at $t = 3$?

A $\dfrac{19}{2}$

B $\dfrac{13}{2}$

C 7

D 8

E $\dfrac{17}{2}$

F 10

Worked solution · A

The area for the first two minutes is $2(3 + 9)/2 = 12$. If the missing rate is r, the last two one-minute trapezia give $(9 + r)/2 + (r + 5)/2 = 7 + r$. Thus $19 + r = 28.5$ and $r = 9.5$ litres per minute.

Q13

Cubic Graphs & Parameters

Answer D

Question 13

Find the complete set of values of k for which

$$2x^3 - 3x^2 - 36x + k = 0$$

has three distinct real solutions.

A $k < 44$

B $k < 81$

C $-81 < k < 44$

D $-44 < k < 81$

E $k > 44$

F $k > 81$

WORKED SOLUTION

Let

$$h(x) = 2x^3 - 3x^2 - 36x.$$

Then

$$h'(x) = 6x^2 - 6x - 36 = 6(x + 2)(x - 3),$$

so the turning points occur at $x = -2$ and $x = 3$. Their values are

$$h(-2) = 44, \qquad h(3) = -81.$$

The original equation is $h(x) = -k$. It has three distinct real roots exactly when the horizontal line $y = -k$ lies strictly between the turning-point values:

$$-81 < -k < 44.$$

Multiplying by -1 reverses the inequalities:

$$-44 < k < 81.$$

The answer is $\boxed{\mathbf{D}}$.

WORKED SOLUTIONS

Advanced integration and calculus reasoning

Questions 14–20 · initial conditions, modulus integrals, symmetry, differential equations, constrained optimisation, rational indices and the Fundamental Theorem of Calculus.

Q14

Differential Equations

Answer B

Question 14

A function y satisfies

$$\frac{dy}{dx} = 7 + \frac{3}{x}, \qquad x > 0.$$

Its general solution is written as

$$y = 7x + 3\ln x + C.$$

Given that $y = 5$ when $x = 3$, find C.

A $16 + 3\ln 3$

B $-16 - 3\ln 3$

C $-2 - 3\ln 3$

D $-16 + 3\ln 3$

E $21 + 3\ln 3$

F 5

WORKED SOLUTION

Substitute the initial condition $(x, y) = (3, 5)$ into

$$y = 7x + 3\ln x + C.$$

This gives

$$5 = 21 + 3\ln 3 + C.$$

Hence

$$C = 5 - 21 - 3\ln 3 = -16 - 3\ln 3.$$

The answer is $\boxed{\text{B}}$.

Q15

Modulus & Integration

Answer D

Question 15

Find the value of

$$\int_0^2 |x-1|\left(3\sqrt{x}-x\sqrt{x}\right)dx.$$

A $\frac{48-4\sqrt{2}}{35}$

B $\frac{4\sqrt{2}}{35}$

C $\frac{24}{35}$

D $\frac{48+4\sqrt{2}}{35}$

E $\frac{24+4\sqrt{2}}{35}$

F $\frac{24-4\sqrt{2}}{35}$

WORKED SOLUTION

Split the integral at $x=1$:

$$I=\int_0^1 (1-x)(3x^{1/2}-x^{3/2})dx+\int_1^2 (x-1)(3x^{1/2}-x^{3/2})dx.$$

Let

$$G(x)=(x-1)(3x^{1/2}-x^{3/2}).$$

An antiderivative of G is

$$F(x)=-\frac{2}{7}x^{7/2}+\frac{8}{5}x^{5/2}-2x^{3/2}.$$

The first integrand is $-G(x)$, so

$$I=-[F(x)]_0^1+[F(x)]_1^2=F(2)-2F(1).$$

Now

$$F(1)=-\frac{24}{35},\qquad F(2)=\frac{4\sqrt{2}}{35}.$$

Therefore

$$I=\frac{4\sqrt{2}}{35}+\frac{48}{35}=\frac{48+4\sqrt{2}}{35}.$$

The answer is $\boxed{\text{D}}$.

Q16

Symmetry & Definite Integrals

Answer F

Question 16

For a function f,

$$\left(\int_0^2 f(x)\,dx\right)\left(\int_0^1 f(x)\,dx + 4\right) = 24.$$

Also, $f(x+1) = f(1-x)$. Which of the following could be the value of

$$\int_1^2 f(x)\,dx?$$

I. -6
II. 8
III. 2

A None

B I only

C II only

D III only

E I & II only

F I & III only

G II & III only

H All three

WORKED SOLUTION

Let

$$A = \int_0^1 f(x)\,dx, \qquad B = \int_1^2 f(x)\,dx.$$

Using $f(x+1) = f(1-x)$ and integrating for $0 \le x \le 1$,

$$B = \int_0^1 f(x+1)dx = \int_0^1 f(1-x)dx = A.$$

Thus $\int_0^2 f = 2A$, and the given condition becomes

$$2A(A+4) = 24.$$

So

$$A^2 + 4A - 12 = 0 \quad \Longrightarrow \quad (A+6)(A-2) = 0.$$

Hence $A = B = -6$ or 2. Statements I and III are possible.

The answer is $\boxed{\mathbf{F}}$.

Q17 Differential Equations & Trigonometry **Answer A**

Question 17

Given that

$$\frac{dy}{dx} = \frac{3}{2}\left(\frac{x^{7/3} - x^{5/3}}{x^{1/3} - 1} + \frac{x^{7/3} - x^{5/3}}{x^{1/3} + 1}\right), \qquad x > 1,$$

and $y = 4$ when $x = 2$, find y when $x = 5$.

- **A** 121
- **B** 117
- **C** 125
- **D** 113
- **E** 129
- **F** 81
- **G** $\frac{243}{2}$

WORKED SOLUTION

Let $u = x^{1/3}$. Then

$$x^{7/3} - x^{5/3} = x^{5/3}(x^{2/3} - 1) = u^5(u - 1)(u + 1).$$

Therefore

$$\frac{x^{7/3} - x^{5/3}}{x^{1/3} - 1} = u^5(u + 1), \qquad \frac{x^{7/3} - x^{5/3}}{x^{1/3} + 1} = u^5(u - 1).$$

Adding the two fractions gives

$$u^5(u + 1) + u^5(u - 1) = 2u^6 = 2x^2.$$

Hence

$$\frac{dy}{dx} = \frac{3}{2}(2x^2) = 3x^2,$$

so

$$y = x^3 + C.$$

Using $y = 4$ when $x = 2$,

$$4 = 8 + C \quad \Longrightarrow \quad C = -4.$$

Thus, when $x = 5$,

$$y = 5^3 - 4 = 125 - 4 = \boxed{121}.$$

The correct answer is A.

Q18 Constrained Integration **Answer D**

Question 18

Given that

$$\int_0^4 (x^3 + ax^2 + bx + 1)dx = 0,$$

where $a \leq 0$ and $b \leq 0$, find the maximum value of

$$\int_0^3 (x^3 + ax^2 + bx + 1)dx.$$

A -15

B 15

C 100

D $-\frac{87}{16}$

E $\frac{123}{2}$

F $-\frac{123}{2}$

WORKED SOLUTION

The first condition gives

$$68 + \frac{64a}{3} + 8b = 0,$$

so

$$b = -\frac{17}{2} - \frac{8a}{3}.$$

Because $b \leq 0$,

$$-\frac{17}{2} - \frac{8a}{3} \leq 0 \quad \Longrightarrow \quad a \geq -\frac{51}{16}.$$

Together with $a \leq 0$, this gives

$$-\frac{51}{16} \leq a \leq 0.$$

Now

$$\int_0^3 (x^3 + ax^2 + bx + 1)dx = \frac{93}{4} + 9a + \frac{9b}{2}$$
$$= -15 - 3a.$$

This is largest when a is smallest, namely $a = -51/16$. Therefore the maximum is

$$-15 + \frac{153}{16} = -\frac{87}{16}.$$

The answer is $\boxed{\text{D}}$.

Q19 Rational Indices & Differentiation **Answer D**

Question 19

Let

$$f(x) = \frac{\left(x^2 - \frac{1}{x}\right)^2}{x^{1/3}}.$$

It is also given that

$$f'(x) = \frac{x^{2/3}}{3}(ax^p + bx^q + cx^r).$$

Find $abc + pqr$.

A 388 | B $\frac{520}{3}$

C 288 | D 316

E 300 | F -80

WORKED SOLUTION

First expand and simplify:

$$f(x) = \frac{x^4 - 2x + x^{-2}}{x^{1/3}}$$
$$= x^{11/3} - 2x^{2/3} + x^{-7/3}.$$

Differentiate:

$$f'(x) = \frac{11}{3}x^{8/3} - \frac{4}{3}x^{-1/3} - \frac{7}{3}x^{-10/3}.$$

Factor out $x^{2/3}/3$:

$$f'(x) = \frac{x^{2/3}}{3}\left(11x^2 - 4x^{-1} - 7x^{-4}\right).$$

Thus

$$a = 11,\ b = -4,\ c = -7, \qquad p = 2,\ q = -1,\ r = -4.$$

Therefore

$$abc + pqr = 11(-4)(-7) + 2(-1)(-4) = 308 + 8 = 316.$$

The answer is $\boxed{\text{D}}$.

Q20 Fundamental Theorem & Modulus **Answer E**

Question 20

Consider

$$g(x) = \int_0^x (t - p) f(t)\, dt, \qquad 0 \le p \le 7.$$

Find

$$\int_0^7 |x - p| f(x)\, dx.$$

A $g(7) - g(0)$

B $g(7) - g(p)$

C $g(0) - p\,g(7)$

D $2g(0) - g(7) + g(p)$

E $g(7) + g(0) - 2g(p)$

F $g(p) + g(7) - 2g(0)$

WORKED SOLUTION

By the Fundamental Theorem of Calculus,

$$g'(x) = (x - p) f(x).$$

Split the modulus integral at $x = p$:

$$\begin{aligned}
\int_0^7 |x - p| f(x) dx &= \int_0^p (p - x) f(x) dx + \int_p^7 (x - p) f(x) dx \\
&= -\int_0^p (x - p) f(x) dx + \int_p^7 (x - p) f(x) dx \\
&= -[g(p) - g(0)] + [g(7) - g(p)] \\
&= g(7) + g(0) - 2g(p).
\end{aligned}$$

The answer is $\boxed{\mathbf{E}}$.